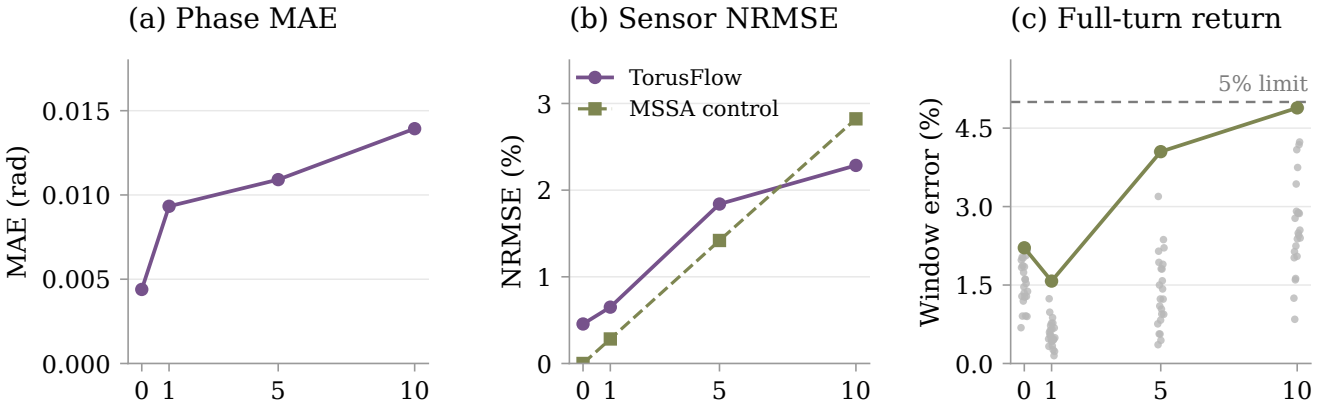
(a) Phase MAE
MAE (rad)
0.015
0.010
0.005
0.000
0
1
5
10
(b) Sensor NRMSE
NRMSE (%)
TorusFlow
MSSA control
3
2
1
0
0
1
5
10
(c) Full-turn return
Window error (%)
5% limit
4.5
3.0
1.5
0.0
0
1
5
10
Noise std / clean TRAIN channel std (%)